

Yashwanth Kumar A

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AI/GenAI Engineer | Agentic AI Systems | RAG, Evaluation, Data Operations and Workflow Automation

Summary

Postgraduate AI/GenAI engineer with **1 year of production experience** evaluating LLM chatbots, RAG pipelines, and agentic AI systems, plus portfolio-grade builds spanning SEC research, experimentation infrastructure, and regulated financial workflows. Hands-on with **Python, SQL, LangGraph, LangChain, FastAPI, RAG, vector retrieval, LLM provider APIs, CI/CD, and evaluation frameworks**; delivered **70%** lower validation effort, **45%** stronger failure detection, and **500+** structured AI quality reports.

Skills

AI/GenAI Engineering: Python, SQL, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Prompt Engineering, RAG Architectures, Hybrid Retrieval, Vector Search, Embeddings, Tool-Calling, Function Orchestration, LLM Provider APIs, Anthropic, Google GenAI, Groq

Backend, Data and Deployment: FastAPI, RESTful APIs, Scalable Backend Services, SQLAlchemy, PostgreSQL, ChromaDB, Redis, dbt-core, SQLite, Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Streamlit

Evaluation, Monitoring and Analytics: RAGAS, DeepEval, Langfuse, LangSmith, LLM Evaluation, Monitoring Frameworks, Cost Tracking, Latency Tracking, Responsible AI, Guardrails, Bayesian A/B Testing, CUPED Variance Reduction

Experience

AramAlgorithm.ai

Mar 2025 – Feb 2026

AI Red Team Tester – GenAI Evaluation, Agentic AI Reliability and Responsible AI

Remote

- **Optimized manual AI safety validation effort by 70%** by designing **500+** Python-based adversarial prompt scenarios across LLM chatbots, RAG pipelines, and agentic AI systems using Garak, PyRIT, and PromptBench, resulting in a **100% adversarial safety pass rate** on 20/20 red-team prompts.
- **Drove a 45% improvement in failure-mode detection** by evaluating grounded retrieval behavior, multi-agent handoffs, orchestration edge cases, and unsafe output patterns, resulting in faster engineering triage and stronger reliability for production-facing GenAI workflows.
- **Delivered 500+ structured evaluation reports** with severity ratings, reproduction steps, impact assessments, and guardrail recommendations for engineering and compliance stakeholders, resulting in clearer remediation priorities and repeatable quality assurance practices across an **11-month** remote workstream.

Projects

CortexAgent – Production Agentic RAG Platform for SEC 10-K Research | GitHub

- **Built a 4-agent LangGraph RAG platform** over **932 SEC filing chunks** using section-aware chunking, BM25 + ChromaDB hybrid retrieval, Reciprocal Rank Fusion, BGE reranking, and auditable chunk IDs, resulting in traceable financial answers grounded in filing evidence.
- **Improved correctness by 59% and faithfulness by 28%** by implementing RAGAS-gated CI/CD release checks, critic-driven revision loops, and a 20-prompt red-team safety suite, resulting in **100% high-severity failure-free responses**.
- **Optimized reliability and unit economics** through Anthropic, Google GenAI, and Groq fallback routing plus API-exposed cost tracking, achieving **\$0.05–\$0.15/query**, **250 ms p50 latency**, and **70% lower demo-failure risk** versus single-provider routing.

ExperimentOS – Product Experimentation and SQL Metric Governance Platform | Live

- **Engineered Bayesian A/B testing and CUPED variance reduction** on **541,909 e-commerce transactions**, reducing required sample size from **8,400 to 5,764 users** and saving **5.3 days** of experiment runtime per test.
- **Implemented dbt-sqlite SQL transformation and metric governance** with a GitHub Actions approval gate, **27 pytest tests**, and LLM-powered decision memos, resulting in **100% metric-change governance** and eliminating **1–2 hours** of manual experiment documentation per test.

Trusted Agentic Banking and Fintech Operating System – Policy-Governed Workflow Control Layer | In Progress

- **Designing a policy-governed Agentic AI control layer** for KYC clearance and insurance claims across **5 trust modules**: Evidence Locker, Privacy Hub, Human Verification Room, Clearance Ledger, and Policy Graph, resulting in auditable, evidence-backed workflow blueprints for regulated finance.
- **Defining reusable orchestration patterns** for permissions, human escalation, final clearance, and replayable audit trails, creating adoption criteria for traceability, least-privilege access, and responsible enterprise automation.

Education

Vellore Institute of Technology

Jun 2020 – Sep 2025

Integrated M.Tech in Computer Science and Engineering, Business Analytics

Chennai, India

- GPA: **8.03/10.0** | Coursework: Machine Learning, NLP, Data Mining, Statistics, Algorithms, Business Analytics

Certifications and Programs

McKinsey.org Forward Program – Selected Learner | LangChain and Agentic AI – DeepLearning.AI | AWS Cloud Essentials | Cisco Data Analytics Essentials